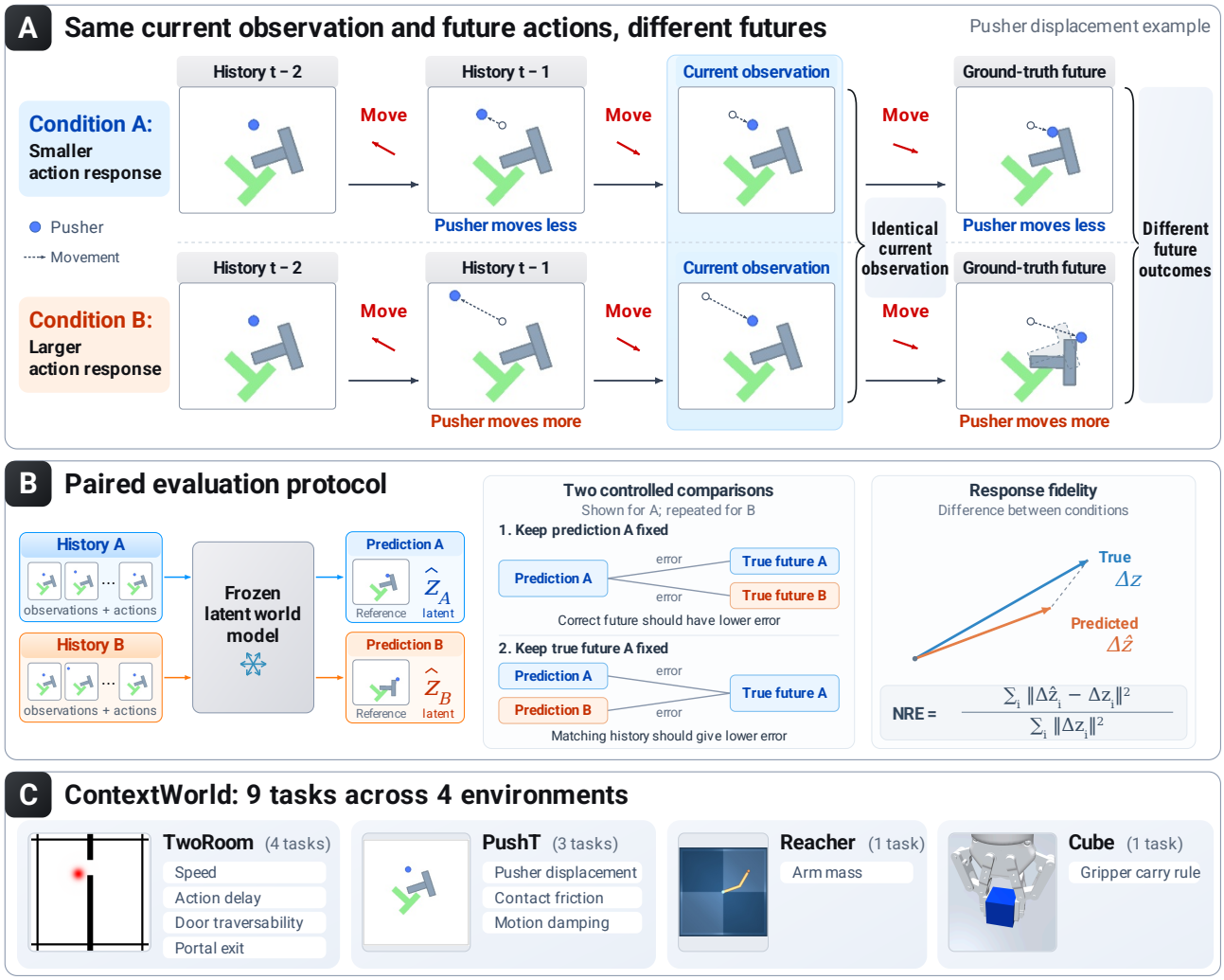
**Layout alternative: visual references beside latent predictions**




**ContextWorld overview.** (A) A recorded pusher-displacement pair. Shared actions reveal different movement distances in the history, return both conditions to the same current observation, and produce different future states. Red arrows summarize action directions; dashed arrows show observed endpoint displacement, including the return movement. The translucent outline in the lower future panel marks the T block's current pose. These overlays are annotations, not model inputs. All images use the same crop of the original observations. (B) A frozen model predicts a future from each history and the shared future actions. The thumbnails labeled Reference show the corresponding true futures for visual context; they are not decoded model predictions. The predicted outputs remain the latent states $\hat{z}_A$ and $\hat{z}_B$. The two comparisons hold different quantities fixed. Future discrimination compares one prediction with the two true futures. History benefit compares the predictions from histories A and B with the same true future. The diagrams show condition A; both comparisons are repeated for B. Connecting lines denote latent prediction-error comparisons, and the text below each diagram states the desired outcome. Response fidelity compares the predicted difference $\Delta\hat{z} = \hat{z}^B - \hat{z}^A$ with the true difference $\Delta z = z^B - z^A$. NRE measures their discrepancy, with zero indicating an exact response. The arrows illustrate this comparison schematically; they are not a projection of measured model outputs. (C) Nine tasks across four environments.